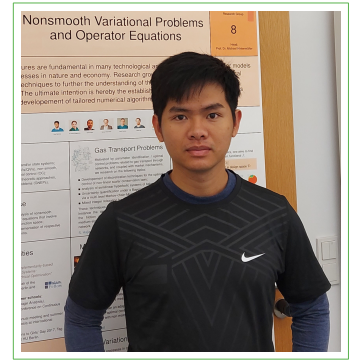


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[Google Scholar/Hong Quan Ba Nguyen](https://scholar.google.com/citations?user=HONGQUANBA)
[ResearchGate/Hong Quan Ba Nguyen](https://www.researchgate.net/profile/Hong-Quan-Ba-Nguyen)



Hong Nguyen

Curriculum Vitae

"The essence of mathematics lies in its freedom." - Georg Cantor

"It is true that a mathematician who is not somewhat of a poet, will never be a perfect mathematician." - Karl Weierstrass

Profile

Applied Mathematician specializing in continuous physical systems (PDEs, fluid dynamics) and mathematical optimization. Highly motivated to bridge rigorous numerical analysis with data-driven operator learning to model and control complex nonlinear dynamical systems. Proven research experience in reduced-order modeling (ROMSOC) and independent theoretical research.

Education

- 2020–2021 **PhD student** enrolled in **Humboldt-Universität zu Berlin**, *Weierstrass Institute for Applied Analysis & Stochastics* (WIAS), Berlin, Germany.
Within the European Innovative Training Network **ROMSOC** (*Reduced Order Modelling, Simulation & Optimization of Coupled Systems*). Focused on shape optimization, fluid mechanics (Navier-Stokes), and reduced-order modeling for continuous physical systems.
- 2018–2019 **Master de Sciences, Technologie, Santé, Mathématiques et Applications**, *Université de Rennes 1*, Rennes, France, *assez bien*.
Master 2 in Fundamental Mathematics & Applications within ENS Rennes program (skipped Master 1).
- 2014–2018 **Bachelor of Science in Mathematics**, *University of Science* (HCMUS), *Viet Nam National University, Ho Chi Minh City* (VNU HCM), Vietnam, *very good*.
Specialized in Mathematical Analysis & Numerical Analysis.
- 2011–2014 **Elementary & Olympiad Mathematics**, *Ben Tre High School for Gifted Students* (CBT), Vietnam, *very good*.

Master Thesis

- Title *Staggered & well-balanced discretization of shallow-water equations.* [[pdf](#)][[slide](#)]
- Supervisor Prof. **Nicolas Seguin**
- Description This thesis investigates some staggered well-balanced finite volume schemes for the shallow water equations. A new staggered scheme, called “offset equilibrium”, is developed to conquer the *dry-wet transition* issues where almost all the traditional finite volume schemes collapse. Higher-order extensions for this staggered scheme are also mentioned.

Master Courses

- 2018–2019 **Courses of Master 2 Program in Fundamental Mathematics with Applications, 2018–2019:**
- Sep 2018 Prof. **Nicoletta Tchou**. *Sobolev spaces & elliptic equation.*
- Sep 2018 Prof. **Florian Méhats**. *Hyperbolic equations.*
- Nov 2018 Prof. **Nicolas Seguin** & **Eric Darrigrand**. *Finite Element Method (FEM).*
- Nov 2018 Prof. **Nicolas Seguin**. *Numerics of transport.*
- Nov 2018 Prof. **Mohammed Lemou**. *Numerical methods for kinetic transport equations.*
- Dec 2018 Prof. **Luis Miguel Rodrigues**. *Master 2 Seminar*
▷ Hong Nguyen. “On the smallest constant for a Gagliardo-Nirenberg functional inequality”. [[pdf](#)][[slide](#)]
- Mar 2019 **Vincent Duchêne**. *Modeling & mathematical analysis of shallow free-surface fluid flows.*
- Mar 2019 Prof. **Marc Quincampoix**. *Optimal control & Hamilton-Jacobi equations.*

Research Interests

- **Nonlinear Dynamical Systems** ▷ Spectral Submanifolds (SSM), Fluid-Structure Interactions, Model Reduction.
- **Data-Driven Modeling & Control** ▷ Operator Learning, Scientific Machine Learning (SciML), Physics-Informed Neural Networks (PINNs).
- **Mathematical Analysis & Fluid Mechanics** ▷ Navier–Stokes equations, Computational Fluid Dynamics (CFDs), Turbulence modeling.
- **Mathematical Optimization** ▷ Optimal Control, Shape & Topology Optimization.
- **Numerical Analysis** ▷ Finite Difference Methods (FDMs), Finite Element Methods (FEMs), Finite Volume Methods (FVMs).

Research & Professional Experience

Vocational & Research

- Jan 2025–now **Lecturer**, *University of Management & Technology (UMT HCMC)*, Ho Chi Minh City.
School of Technology (SOT): expertise: Mathematics for Data Science & Artificial Intelligence.
- Jan 2025–now **Mathematical Olympiad trainer**, *UMT HCMC*, Ho Chi Minh City.
Training undergraduate students for Vietnamese Mathematical Competitions for Undergraduate Students.
- Jan 2025–now **Olympiad in Informatics trainer**, *UMT HCMC*, Ho Chi Minh City.
Training undergraduate students for Vietnamese Olympiad in Informatics for Undergraduate Students.
- Jan 2022–Dec 2024 **Independent Researcher**, Ben Tre city, Vietnam.
Conducted independent research focusing on applied mathematics, optimization, and algorithmic reduction. Authored and submitted multiple preprints bridging theoretical mathematics with computational applications.
- Apr 2021–Dec 2021 **Software Development Engineer (SDE)**, *Math+ Tec GmbH*, Vienna, Austria.
Secondment for technical & scientific training as part of the ROMSOC project. Applied model reduction techniques in an industrial context.
- 2020–2021 **Marie Curie PhD Researcher**, *Weierstrass Institute (WIAS)*, Berlin, Germany.
Marie Skłodowska-Curie fellow within the ROMSOC network. Investigated optimal shape design and model reduction of coupled systems.
- Mar–Jun 2019 **Master Internship**, *Institut de recherche mathématique de Rennes – IRMAR*, Rennes, France.

Miscellaneous

- 2015–2017 **Olympiad Trainer**, *Ben Tre High School for Gifted Students (CBT)*, Vietnam.
Trainer of the selected team of CBT for Vietnamese Mathematical Olympiads (VMOs).

Publications

- [math.AP] Nguyen Anh Dao, Jesús Ildefonso Díaz, Quan Ba Hong Nguyen. “*Pointwise Gradient Estimates in Multi-dimensional Slow Diffusion Equations with a Singular Quenching Term*”. *Advances in Nonlinear Studies* (2020). DOI <https://doi.org/10.1515/ans-2020-2076>. [pdf]

Preprints

Under [arXiv Category Taxonomy](#) abbreviations:

Computer Science

- [cs.IT] Nguyen Quan Ba Hong. “*A Functional Level-Set Slack Identity: Elevating Empirical OOD Baselines into Exact Continuous Combinatorial Equations*”. [submitted to IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI); current status: under review]
- [cs.IT] Nguyen Quan Ba Hong. “*Boundary Optimization for Rough Sets: Continuous Integral Decoupling and Intractability Barriers*”. [submitted to European Journal of Operational Research (EJOR)]
- [cs.IT] Nguyen Quan Ba Hong. “*Real-Time Optimization of Streaming Rough Set Boundaries via $\mathcal{O}(1)$ Deamortized Shadow Registers*”. [submitted to Information Science (IS)]
- [cs.CC] Nguyen Quan Ba Hong. “*A Measure-Theoretic Complexity Dichotomy for Combinatorial Up-Set Summations*”. [submitted to SIAM Journal on Computing (SICOMP)]
- [cs.ML] Nguyen Quan Ba Hong, Tran Dan Thu. “*SUAZ-FS: Stochastic Submodular Optimization for Large-Scale Unsupervised Feature Selection*”. [preparing to submit]
- [cs.] Nguyen Quan Ba Hong. “*Global Theoretical Optimality of Exact Multi-Criteria Shortest Paths via Fine-Grained Complexity*”. [preparing]

Mathematics

- [math.EA] Nguyen Quan Ba Hong. “*Algorithmic Reduction of Elementary Symmetric Inequalities via Invariant Positivstellensatz Certificates*”. [submitted to SIAM Journal on Applied Algebra & Geometry (SIAGA)]
- [math.NT] Nguyen Quan Ba Hong. “*An Automated Resolution and Generalization of Rowland’s Binomial Divisibility Conjectures via Algebraic Rank Minimization*”. [submitted to Advances in Mathematics, status: under review]
- [math.AP] Nguyen Quan Ba Hong. “*Universal Gradient Estimates and Exact Free Boundary Profiles for the Singular Porous Medium Equation*”. [submitted to Archive for Rational Mechanics and Analysis (ARMA), status: editor assignment pending] (extending NQBH’s bachelor thesis)
- [math.AP] Nguyen Quan Ba Hong. “*Spatial Decoupling and Complete Quenching for Singular Porous Medium Equations*”. [submitted to Archive for Rational Mechanics and Analysis (ARMA), status: editor assignment pending] (extending NQBH’s bachelor thesis)
- [math.NA/FVM] Nguyen Quan Ba Hong. “*Convergence and Asymptotic Commutativity of the Adjoint PISO Algorithm on Polyhedral Meshes*”. [submitted to SIAM Journal of Numerical Analysis (SINUM)]
- [math.CO] Nguyen Quan Ba Hong. “*Boundary Optimization for General Overlapping Coverings via Extremal Combinatorics*”. [submitted to Information Science]
- [math.CO] Nguyen Quan Ba Hong. “*Algebraic and Analytic Invariants of Finite Alexandrov Topologies: Möbius-Inverted Extremal Identities and Log-Concave Boundary Polynomials*”. [submitted to JCTA]
- [math.MO] Nguyen Quan Ba Hong. “*Exact Explicit Shape Calculus for the Regularized k - ε Turbulence Model on Non-Cylindrical Moving Domains*”. [preparing]
- [math.ML] Nguyen Quan Ba Hong. “*Provably Convergent Anisotropic Graph Neural Diffusion for Exact Anomaly Separation*”. [preparing]
- [math.NA/FVM] Nguyen Quan Ba Hong. “*Global Convergence and Lax–Wendroff Consistency of the Offset Equilibrium Staggered Scheme for Shadow Water Equations at Dry-Wet Transitions*”. [submitted to SIAM Journal on Numerical Analysis (SINUM)] (extending NQBH’s master thesis)
- [math.NA/FVM] Nguyen Quan Ba Hong. “*Möbius Convolutions and Functorial Ahlswede–Zhang Identities over Incidence Algebras*”. [preparing]
- [math.AT] Nguyen Quan Ba Hong. “*Alexandrov Completions, Simplicial Nerves, and Ahlswede–Zhang Obstructions over Finite Covering Spaces*”. [submitted to Journal of Applied and Computational Topology (JACT)] (sequel ↓)
- [cs.AT] Nguyen Quan Ba Hong. “*Homological Quantification of the Čech Defect: Quillen Fibrations and Ahlswede–Zhang Obstructions over Finite Posets*”. [submitted to Journal of Combinatorial Theory, Series A (JCTA)]; current status: with editor/seeking expert advice (foundation ↑)

Pedagogy/Education

- [math.HO] Nguyen Quan Ba Hong. “*Beyond Code Generation: The Cognitive Necessity of Analytic Derivation in Mathematical Olympiad Training in the Era of Generative AI*”. [revising to resubmit]
- [math.HO] Nguyen Quan Ba Hong. “*Fundamental Asymptotic Limits of Generative AI in Discrete Mathematical Reasoning*”. [revising to resubmit]

Physics

- [nlin.CD] Nguyen Quan Ba Hong. “*Topological Phase Transitions and Entropy Bifurcations in Dynamic Equivalence Manifolds*”. [submitted to Chaos, Solitons and Fractals, status: with editor]

Books & Other Writings

- [wrt1] Nguyen Quan Ba Hong. “*Vietnamese National High School Graduation Mathematics Exam & Beyond – Kỳ Thi THPT Quốc Gia Môn Toán của Việt Nam: Trực Giác, Lập Trình, & Tổng Quát Hóa*”. [In preparation]
- [book1] Nguyen Quan Ba Hong. “*Theoretical Competitive Programming (Olympiad in Informatics & ACM-ICPC)*”. [In preparation]
- [book2] Nguyen Quan Ba Hong. “*Programmable Mathematical Olympiads (VMC & High School)*”. [In preparation]
- [book3] Nguyễn Quân Bá Hồng, Nguyễn Quân Trung Nhân. *Some Thoughts on Learning, Teaching, & Research: A Short Novel – Vài Suy Nghĩ Về Việc Học, Việc Dạy, & Nghề Nghiên Cứu: 1 Tiểu Thuyết Ngắn*.

Courses

- Oct 12–23, 2020 Thematic Einstein Semester (TES): *Energy-based mathematical methods for reactive multiphase flows*.
▷ TES Student Compact Course: *Variational Methods for Fluids & Solids*.

Apr–Jun 2020 Dr. [Stephan Schmidt](#). *Computational Geometry & Shape Optimization*. Master course, Humboldt Universität zu Berlin, 2020.

Seminars

- Mar 2020–now [Berliner Oberseminar Nichtlineare partielle Differentialgleichungen \(Langenbach-Seminar\)](#), Research Group 1: *Partial Differential Equations*, WIAS Berlin, Germany.
- Mar 2020–now [Material Modeling Seminar](#), WIAS Berlin, Germany.
- Mar 2020–now [Numerical Mathematics Seminars](#), Research Group 3: *Numerical Mathematics & Scientific Computing*, WIAS Berlin, Germany.
- Mar 2020–now [Modern Methods in Applied Stochastics & Nonparametric Statistics](#), Research Group 6: *Stochastic Algorithms & Nonparametric Statistics*, WIAS Berlin, Germany.
- Mar 2020–now [Joint Research Seminar on Mathematical Optimization](#), Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.
- Mar 2021–now [Optimization, Control & Inverse Problems](#), Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.
- Mar 2020–now [Berlin Mathematics Research Center MATH⁺ Events](#).
- Nov 2020–Apr 2021 ROMSOC Scientific ESRs Seminar.

Schools

- Jul 26–30, 2021 [Summer Schools on Applied Mathematics and Mechanics 2021 \(SAMM 2021\)](#): *Shape & Topology Optimization*. [GAMM Juniors](#), Graz University of Technology (Technische Universität Graz), Graz, Austria.
- Jul 1–2, 2021 [Durham Days of Analysis & PDE Summer School](#). Durham University, United Kingdom.
- Sep 7–10, 2020 [Model Order Reduction Summer School 2020 \(MORSS2020\)](#). École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.
- Jul 20–24, 2020 [162nd European Study Group with Industry](#). University of Leeds, United Kingdom.

Workshops

- Apr 29, 2021 [FUHRI2021](#): Finite volUme metHods for Real-world appLications, WIAS Berlin, Germany.
- Sep 28–29, 2020 ROMSOC Internal Workshop.
- Sep 15, 2020 ROMSOC Ethical Monitoring - 3rd Workshop on Ethics.

Conferences

- Mar 24–25, 2021 Annual Meeting of the Second Phase of the DFG Priority Programme 1962 ([SPP1962](#)): *Non-smooth & Complementarity-based Distributed Parameter Systems: Simulation & Hierarchical Optimization*. Deutsche Forschungsgemeinschaft (DFG) + WIAS, Germany.
- Sep 14–17, 2020 [Deutsche Mathematiker-Vereinigung](#) (German Mathematical Society) Annual Meeting 2020 ([DMV2020](#)), Technische Universität Chemnitz, Chemnitz, Germany.

Talks

- Apr 12, 2021 Hồng Quân Bá Nguyễn (WIAS Berlin). “*Optimal Shape Design of Air Ducts in Combustion Engines: Design a General Shape Optimization Framework*”. ROMSOC ESR Seminar. [[abstract](#)]
- Sep 7, 2020 Hong Nguyen (WIAS Berlin). “*A shape optimization problem for stationary Navier-Stokes flows in three-dimensional tubes*”. [Model Order Reduction Summer School 2020 \(MORSS2020\)](#), École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland. [[abstract](#)][[slide](#)]
- Jun 26, 2019 Hong Nguyen. “*Staggered & well-balanced discretization of shallow water equations*”. PhD interview, Department of Mathematics, RWTH Aachen University, Aachen, Germany. [[slide](#)]

Miscellaneous

- Feb 22, 2021 WIAS-Day2021 ▷ [Hong Quan Ba Nguyen \(Research Group 8\)'s Research Profile](#).
- Nov 30, 2020 [ROMSOC Blog](#) ▷ Hong Nguyen. *Optimal shape design of air ducts* [[pdf](#)].
- Jun 18, 2020 [Prof. Luis Miguel Rodrigues's recognition of Master equivalence & admission to doctoral studies](#) of NGUYEN Quan Ba Hong. IRMAR, Rennes, France.
- May 27, 2019 [Prof. Luis Miguel Rodrigues's recommendation letter](#) for NGUYEN Quan Ba Hong. IRMAR, Rennes, France.

Visits

- Apr 2021–Nov, 2021 [MATH+TEC GmbH](#), Vienna, Austria.
Apr 2021–Nov 2021 [Vienna University of Technology/Technische Universität Wien](#), Vienna, Austria.
Apr 23, 2021 [University of Vienna/Universität Wien](#), Vienna, Austria.
Mar 2020–Nov 2021 [Humboldt-Universität zu Berlin](#), Berlin, Germany.
Jun 25–27, 2019 [Department of Mathematics, Rheinisch-Westfälische Technische Hochschule \(RWTH\) Aachen University](#), Aachen, Germany.
Sep 2018–Jun 2019 [UFR Mathématiques, Université de Rennes 1](#), Rennes, Bretagne, France.
Sep 2018–Dec 2018 [Université Rennes 2 - UFR Sciences sociales](#), Rennes, Bretagne, France.
Mar 2017 [Vietnam Institute for Advanced Studies in Mathematics \(VIASM\)](#), Hanoi, Vietnam.

Awards & Scholarships

- Mar 2020–Aug 2021 [Marie Skłodowska-Curie Actions \(MSCA\) PhD Fellowship](#), *Horizon 2020 - Innovative Training Network - European Industrial Doctorate Programs*, Grant Agreement number: 765374 - [Reduced Order Modelling, Simulation & Optimization of Coupled Systems \(ROMSOC\)](#) [H2020-MSCA-ITN-2017-EID/ROMSOC/ESR11].
2018–2019 [Lebesgue Master Scholarship](#) for *Master 2* program within ENS Rennes program.
2017 Scholarship of [Vietnam Institute for Advanced Studies in Mathematics \(VIASM\)](#).
2016 Special Prize $\Leftarrow$ First Prize in Algebra + First Prize in Analysis, in [Vietnamese Mathematical Olympiad for College Students \(VMC 2016\)](#).
2015 Second Prize in Algebra, Second Prize in Analysis, in [Vietnamese Mathematical Olympiad for College Students \(VMC 2015\)](#).
2014 First Prize #1 in Ben Tre city Mathematical Olympiad for High School Students.
2013 Second Prize in Ben Tre city Mathematical Olympiad for High School Students.
2012 Gold Medal ranked #2, in [Vietnamese Apr 30th Mathematical Olympiad for Central & Southern Vietnam](#).

Computer Skills

- GitHub [GitHub/NQBH](#) $\triangleright$ [GitHub/ROMSOC project](#).
Basic R, Fortran, Fireshape, Firedrake, FEniCS, STAR-CCM+.
Intermediate Antigravity CLI, git, OpenFOAM, LLMs (Gemini's ecosystem).
Advanced C/C++, UNIX/Linux [openSUSE, (K)Ubuntu], $\text{\TeX/L\TeX}$, MATLAB, Python.

Languages

- Vietnamese Mother tongue
English Intermediate, focus more on reading & writing than speaking

Collaborators

In alphabetical order: $\emptyset$ [filling in]

Teaching

- Mathematics [Discrete Mathematics](#), [Linear Algebra](#), [Introduction to Mathematical Analysis](#), [Combinatorics & Graph Theory](#) (Summer 2025, Summer 2026).
Computer Science [Fundamentals of Programming \(C/C++\)](#), [Information Technology Fundamentals II \(Python\)](#), [Machine Learning with Python](#) (Summer 2025, Summer 2026).
Supervision [Bachelor Thesis Supervision in Mathematics/Computer Science](#) (Spring 2026).

Students

In alphabetical order:

- [1] NGUYỄN NGỌC THẠCH (2026) (Software Engineering). *Chronix: Personalized Scheduling with AI and Chronobiology for Optimal Productivity – Chronix: Hệ Thống Lập Lịch Cá Nhân Hoá Tối Ưu Năng Suất Dựa Trên AI và Nhịp Điều Sinh Học*. Current: Software Engineer at Zalo, VNG Corporation, Vietnam.

Referees

In chronological order:

- [1] **Dr. Đào Mạnh Khang**

Bachelor Teacher

[2] **Dr. Nguyễn Tấn Trung**

Bachelor Teacher

[3] **Dr. Đào Nguyên Anh**

Bachelor Supervisor

[6] **Prof. Dr. Trần Đan Thư**

Head of IT Department of UMT

Hobbies

- Psychology Ψ
- Philosophy Φ
- Literary
- Writing, blogging
- Arts (emotions carriers): movies, animes, mangas, (instrumental) musics, etc. ☐

[Updated to Aug 17, 2026]